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# Hard-Budget Feature Selection with Resampled Mutual-Information Repair in EHOA

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Abstract. We evaluate hard-budget repair in EHOA with fold-local mutual-information utility and selected-column imputation, scaling and SMOTE. The five-seed primary analysis has 750 runs on ten datasets; factorial, larger-search and earlier sensitivity experiments bring this study to 1464 runs. Resampled repair has balanced accuracy 0.7932, synthetic cost ratio 0.2774 and Nogueira stability 0.2241; Friedman balanced-accuracy p is 0.14237. Corrected comparisons do not establish predictive superiority. A separately versioned exploratory transfer adds 25 source-only fits on RPDE/DFA/PPE, evaluated on an 80-subject public cohort. Its five-seed resampled ensemble has subject-level balanced accuracy 0.575, with descriptive 95% interval [0.475,0.675]. A published FNA service fee illustrates shared acquisition costs: deleting image-derived columns cannot reduce that shared fee. This neither validates the original 22-feature model nor establishes clinical savings. Public protocols and audited outputs support reproducibility; clinical subject identity, original historical evidence and real marginal feature costs remain unresolved.

Keywords. feature selection, hard budget, resampled mutual information, subset stability

## 1 Problem and contribution

A small feature set need not be inexpensive: acquisition costs can differ between variables. Hard-budget selection limits the sum of selected costs [1]. Subset stability measures how feature choices vary across training samples and random search, rather than how accuracy or convergence varies [2]. In biomedical classification, this distinction matters when investigators interpret a selected measurement panel.

Medical feature selection spans imaging, signals and molecular data [3]. Predictive discrimination, reproducible feature identity and acquisition burden are separate objectives. Correlated descriptors can substitute for one another, so a changing mask need not imply poor prediction; conversely, a repeated mask does not establish a valid biomarker. This motivates reporting balanced accuracy, cost and chance-corrected subset stability together.

We retain the EHOA search [4] and compare unconstrained EHOA with four repairs using the same selected-dimension evaluation pipeline. Among constrained variants, only repair utility differs. FoldSafeEHOA implements fold-local utility and candidate-specific training; these fresh results are separate from the archived CostStableEHOA protocol. The contribution is a controlled, auditable comparison, not a new optimization family.

## 2 Related studies

Stability metrics can reward reproducible yet weak predictors; importance-weighted stability addresses this concern [5]. Medical budget selection with penalized risk [6], sequential acquisition [7], and grouped costs with shadow features [8] solve related but different problems. Table 1 summarizes these distinctions; NR means not reported in the inspected source. Their numerical results are not directly comparable with our protocol.

Table 1. Related-work comparison. Full evaluation details are in the supplement.

| Study | Budget | Costs | Stability | Evaluation | Difference |
|---|---|---|---|---|---|
| Jagdhuber [1] | Yes | Additive | NR | Simulation | Feasible GA |
| Nogueira [2] | NR | NR | Chance-corrected | Resampling | Metric |
| Hamer [5] | NR | NR | Importance-weighted | Resampling | Metric limits |
| Klonecki [6] | Yes | Feature / proxy | NR | Medical data | Penalized ERM |
| Valancius [7] | NR | Sequential | NR | AFA benchmarks | Patient-wise policy |
| Klonecki [8] | Yes | Grouped | NR | MIMIC | Shadow stopping |
| This work | Yes | Synthetic additive | Nogueira | 5 x 3 outer | MI/cost repair |

Classical stability selection aggregates repeated subsample selections and, under specified assumptions and thresholds, controls false selections [9,10]. Our four MI resamples instead estimate a deletion utility before wrapper search. We neither threshold repeated selection probabilities nor inherit those error-control guarantees. Nogueira stability evaluates the final masks [2]; it is not the optimized fitness. Active acquisition [7] chooses measurements sequentially per case, whereas our repair returns a fixed feasible subset.

## 3 Implemented EHOA

Each hiker has a continuous position x and a velocity v; a binary mask z selects features. Initialization uses a tent sequence with initial value 0.7 and parameter 0.4, mapped to [-6,6]. Velocities start uniformly in [-1.2,1.2]. For iteration t of T=10, the implemented update is:

$$v_i \leftarrow clip[-12,12](w_t v_i + SF_t[2r_1(p_i - x_i) + 2r_2(g - x_i)])$$
$$x_i \leftarrow clip[-6,6](x_i + v_i)$$
$$w_t = 0.9 - 0.5t/T;\ SF_t = 3 - 2t/T$$

Here p_i and g are personal and global best continuous positions, and r_1,r_2 are independent per-feature uniform draws. The sigmoid gives selection probability 1/(1+exp(-x_j)); an empty sampled mask restores the largest probability. The implementation keeps continuous leaders separate from repaired masks, caches fitness for duplicate masks, and stops after ten iterations without early stopping.

## 4 Fold-local budget repair and stability

Costs are fixed positive synthetic additive units, drawn with NumPy seed 1000+D from lognormal(0,0.7). The primary budget is B=0.30 sum(c); sensitivity uses 0.15 and 0.50. Costs are independent of labels and do not represent money. Every constrained nonempty mask must satisfy sum(z_j c_j)<=B. Shared acquisition costs require a different grouped model [6,8].

$$u_j = max(0, \bar{I}_j - \lambda s_j);\ q_j = u_j/c_j$$

For each inner fold, MI is fitted to its imputed training rows only. Four class-stratified samples draw ceil(0.8 n_class) rows with replacement (minimum two). The continuous-discrete estimator uses three neighbors [11]; Dermatology treats its first 33 categorical/ordinal inputs as discrete and age as continuous. MI is in nats and s_j is sample SD. Lambda=1 gives a conservative utility, not a calibrated confidence bound; there is no division by sqrt(R). Final outer-training utility is separate and never used to repair inner-validation candidates.

Table 2. Four repairs share the same deletion rule.

| Repair | Utility u_j | Meaning |
|---|---|---|
| Random/cost | U(0,1), fixed per training fold | Not uniform deletion |
| Cheapest | 1 | Largest cost first |
| MI/cost | Single-sample MI | Relevance per cost |
| Resampled/cost | max(0, mean MI - SD) | Conservative heuristic |

Search samples a raw mask. Each inner fold repairs this candidate with its own training utility, then fits imputation, scaling and SMOTE on the selected columns [12]. Validation rows are only transformed and predicted. Fitness averages ordinary accuracy and selected size across three independently repaired folds. The winning raw candidate is finally repaired with outer-training utility; inner and final masks may differ. The outer classifier is refitted using only final selected columns. This evaluates a complete selection-and-repair procedure, not a fixed mask reused across all inner folds [13].

$$F(z) = mean_f\ F_f(z)$$
$$F_f(z) = 0.99(1 - Acc_f) + 0.01k_f/D$$

Deletion removes the active feature minimizing utility/cost while over budget. An unaffordable singleton is cleared; an empty mask restores the affordable highest-ratio feature. Positive costs and finite deletion ensure feasibility when one feature is affordable, as verified here. This guarantees budget compliance, not optimal prediction. MI remains marginal and can overlook interactions or redundancy.

$$s^2_j = M/(M-1) \cdot p_j(1-p_j)$$
$$\Phi = clip[-1,1](1 - mean(s^2_j)/[(\bar{k}/D)(1-\bar{k}/D)])$$

Nogueira stability [2] uses M=15 final masks per dataset/configuration, selection frequencies p_j and mean size k-bar. Identical full/empty masks return one in the code. The masks overlap in training data and are dependent; their copied stability values are not fifteen independent observations. Jaccard is a supplementary, uncorrected overlap measure.

## 5 Datasets and source corrections

Table 3. Fresh-run dataset specifications; U links to archive.ics.uci.edu/dataset/{id}.

| Dataset | N | D | K | Class counts | Source |
|---|---|---|---|---|---|
| Breast Cancer | 569 | 30 | 2 | 212/ 357 | U17 |
| Wine | 178 | 13 | 3 | 59/ 71/ 48 | U109 |
| Ionosphere | 350 | 34 | 2 | 125/ 225 | U52 |
| Sonar | 208 | 60 | 2 | 111/ 97 | U151 |
| Dermatology | 366 | 34 | 6 | 112/ 61/ 72/ 49/ 52/ 20 | U33 |
| Spambase | 4210 | 57 | 2 | 2531/ 1679 | U94 |
| Parkinson (grouped) | 195 | 22 | 2 | 48/ 147 | U174 |
| Hill Valley | 1212 | 100 | 2 | 606/ 606 | U166 |
| Steel (corrected) | 1941 | 27 | 7 | 158/ 190/ 391/ 72/ 55/ 402/ 673 | U198 |
| Madelon | 2600 | 500 | 2 | 1300/ 1300 | U171 |

Exact (X,y) duplicates are removed: Ionosphere uses 350 rows and Spambase 4210. Steel removes six target-indicator inputs and reconstructs seven fault classes. Parkinson recordings match the UCI numeric file and use 32 subject prefixes in both outer and inner group splits. The UCI narrative states 31 participants, so clinical group identity remains unverified. Original raw data and earlier results are preserved.

## 6 Fresh protocol and statistical analysis

The first fold-safe study used three seeds; two additional seeds give five (72001-72005), three outer folds and 750 primary runs across ten datasets and five methods. The extension protocol and code hashes were published in GitHub commit 72c87b6 before its 690 runs: 300 seed replications, 240 complete factorial runs and 150 larger-search runs. Earlier outcomes were known. This is prospective versioning, not an independent formal preregistration or registration of past experiments. Inner utility uses training labels only; outer tests are never used to choose a parameter winner.

Table 4. Actual frozen settings.

| Parameter | Value |
|---|---|
| Population / iterations | Primary 8/10; search study 16/20 |
| Position / velocity clip | [-6,6] / [-12,12] |
| Tent / sweep / inertia | 0.7,p=0.4 / 3 to 1 / 0.9 to 0.4 |
| c1,c2 / alpha | 2,2 / 0.99 |
| Outer / inner folds | 3 / 3; Parkinson grouped |
| Outer seeds | 72001-72005 |
| Selector seed | outer seed + fold index x 100003 |
| Primary R / fraction / lambda | 4 / 0.8 with replacement / 1 |
| Sensitivity R / lambda | Factorial 4,8 / 0,2; earlier OFAT 4,8,16 / 0,1,2 |
| Cost / budgets | lognormal(0,0.7) / 15,30,50% |
| Classifier | 5-NN; uniform; Euclidean |
| Imputation / scale / SMOTE | Training-selected columns only |
| Timing | Includes utility and final classification |

Each dataset supplies one mean per method and one stability estimate [14]. Friedman tests use ten dataset blocks, tie correction and df=4. Two-sided Wilcoxon compares resampled repair with four controls, followed by Holm within each metric and across all 24 contrasts. Six omnibus tests are also corrected. Dataset-bootstrap intervals use 20,000 draws (seed 20261012) and are descriptive and unadjusted. Fifteen dependent masks are never treated as independent replicates.

## 7 Results and sensitivity

Table 5. Equal-weight means over ten datasets; V is budget exceedance.

| Method | BA | Cost ratio | Size | Stability | V |
|---|---|---|---|---|---|
| EHOA | 0.7946 | 0.4947 | 42.84 | 0.1036 | 145/150 |
| Random/cost | 0.7814 | 0.2724 | 31.59 | 0.1217 | 0/150 |
| Cheapest | 0.7944 | 0.2765 | 35.05 | 0.2476 | 0/150 |
| MI/cost | 0.7949 | 0.2787 | 27.31 | 0.2203 | 0/150 |
| Resamp./cost | 0.7932 | 0.2774 | 31.24 | 0.2241 | 0/150 |

Table 6. Friedman and mean ranks in E/R/C/M/S order; lower rank is better.

| Metric | Chi2 | p | E | R | C | M | S |
|---|---|---|---|---|---|---|---|
| BA | 6.8800 | 0.14237 | 2.50 | 4.10 | 2.60 | 2.70 | 3.10 |
| Cost | 21.2000 | 0.00029 | 5.00 | 2.20 | 2.60 | 2.90 | 2.30 |
| Size | 29.1313 | <0.00001 | 5.00 | 1.85 | 3.55 | 1.75 | 2.85 |
| Stability | 29.6800 | <0.00001 | 4.70 | 4.20 | 1.60 | 2.20 | 2.30 |

Resampled minus EHOA changes BA by -0.00141, with W=22, within-metric Holm p=1.00000, rank-biserial r=-0.200 and descriptive 95% interval [-0.01024,0.00799]. Stability changes by +0.12045 (Holm within metric 0.00781; all 24 contrasts 0.04688). Cost changes by -0.21732 (Holm within metric 0.00781; all 24 0.04688). These contrasts include imposing a budget on unconstrained EHOA, and do not isolate a resampling benefit. Nonsignificance does not prove equivalence.

The complete 2x2x2 factorial crosses budget 0.15/0.50, R=4/8 and lambda=0/2 on all ten datasets. Orthogonal contrasts estimate main, pairwise and three-way endpoint effects. Each cell has one seed and three outer folds, so interactions are descriptive and do not cover the full continuous parameter space. A paired larger-search study compares 16 hikers and 20 iterations with 8 and 10 using seed 72001, shared outer splits and all five methods. All configurations were reported; none was selected from outer-test results.

Cheapest repair has higher mean stability (0.2476) than resampled repair (0.2241); the paired delta is -0.02347, with within-metric Holm p=0.21094. Resampling therefore has no corrected stability advantage over Cheapest or single-MI repair. A descriptive improvement over an unconstrained baseline is not evidence of superiority over feasible simpler alternatives.

The endpoint R=8 versus R=4 balanced-accuracy effect is +0.00786, averaged over budgets and penalties, with descriptive interval [0.00154,0.01527]. Larger search changes resampled inner fitness by -0.01554, but its outer BA delta is -0.00424, interval [-0.02869,0.01940]. Thus better optimized inner fitness does not establish better held-out prediction. These intervals are unadjusted descriptive contrasts; they do not select the primary setting.

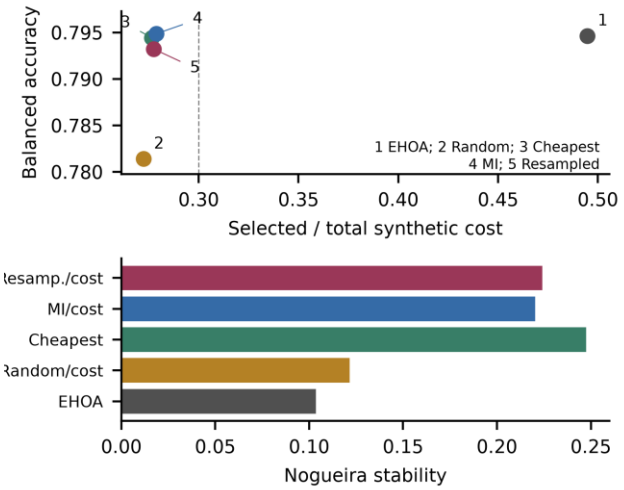



Figure 1. Fresh ten-dataset means; synthetic cost and descriptive subset stability.

## 8 Biomedical case and congress relevance

Breast Cancer Wisconsin (Diagnostic) has 569 fine-needle-aspirate cases, 212 malignant and 357 benign, with 30 nuclear-image descriptors [15]. Table 7 and Figure 2 show fifteen masks per method. Counts describe computational selections, not causal biomarkers. Several descriptors share the same image acquisition, so additive synthetic costs cannot be interpreted as biopsy-price savings. Label 0 is malignant and label 1 benign; class-specific recall is explicitly reported.

Table 7. Breast-cancer case: fifteen runs per method.

| Method | BA | Mal. sens. | Ben. spec. | Cost | Stability |
|---|---|---|---|---|---|
| EHOA | 0.9512 | 0.9415 | 0.9608 | 0.5620 | 0.0727 |
| Random/cost | 0.9382 | 0.9217 | 0.9546 | 0.2525 | 0.0988 |
| Cheapest | 0.9519 | 0.9368 | 0.9669 | 0.2707 | 0.1975 |
| MI/cost | 0.9517 | 0.9387 | 0.9647 | 0.2771 | 0.2046 |
| Resamp./cost | 0.9493 | 0.9340 | 0.9647 | 0.2751 | 0.1757 |

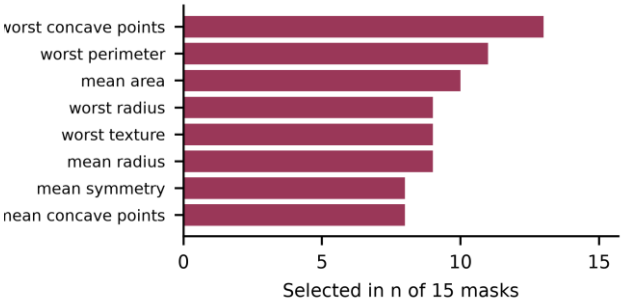



Figure 2. Top eight feature selection counts in fifteen resampled-repair masks.

The shared-service scenario uses published Montana FNA interpretation code 88173: global USD 164.34, effective June 30, 2024 [17]. All 30 WDBC columns derive from FNA images [15]; mapping them to one acquisition group is an illustrative assumption, not a validated invoice. All 75 nonempty primary breast-cancer masks retain the full shared fee: zero service-fee savings and no feasible nonempty subset at a 30% group budget. Collection, computation time, risk and marginal charges are unknown.

The exploratory transfer protocol 7f716c5 was published before 25 fits (five methods x five seeds). Training/repair uses Oxford 195 only with inner subject-prefix separation. The separate UCI 489 target has 240 recordings from 80 subjects (40 PD/40 controls) [16]. Only shared RPDE/DFA/PPE enter selection; equal synthetic costs permit one column for repairs, while EHOA is unconstrained. Five-

seed probabilities are averaged, then three recordings per subject; threshold 0.5. Target labels are used only for evaluation. Intervals are unadjusted 10000-resample stratified subject bootstraps conditional on fitted models, not independent seed replications or complete training uncertainty.

Table 8. External three-feature transfer; five-seed subject ensembles. E=EHOA; R/C/M/S are repairs. BA intervals include 0.5; no clinical superiority.

| Methods | BA [95% descriptive] | Sens./Spec. | AUC |
|---|---|---|---|
| E | 0.575 [0.475,0.675] | 0.725/0.425 | 0.5794 |
| R/C/M/S | 0.575 [0.475,0.675] | 0.725/0.425 | 0.5500 |

## 9 Limitations and reproducibility

Finite search cannot prove convergence; factorial and larger-search contrasts use one seed. Marginal MI minus SD has no calibrated coverage or false-selection guarantee. Timings use three/six workers. Original recording identity and missing historical records remain unverified. The three-feature transfer uses matching names, without certified extraction/calibration equivalence; it cannot validate the original 22-feature model or prospective clinical use.

github.com/Amirgh23/coststable-ehoa provides code, attributed public data, all predictions, masks and audits, manuscripts and responses. The original 1464 runs and the 25 transfer fits have separate protocols; they are not pooled statistically. Hill-Valley uses UCI 166/OpenML 1479.

## 10 Conclusion

Fold-local repair enforces synthetic budgets. Corrected results do not show universal predictive superiority; exploratory public-cohort transfer is weak and shared-service fees yield no savings from column deletion. Measured marginal costs and clinical validation are required for deployment.

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